

**EMPIRICAL PAPER II**  
**THE REQUIREMENT–CAPACITY RACE IN NORTH AMERICAN**  
**ELECTRICITY**

**A Hostile-Domain Test of the SignalRupture Structural Grammar**

SignalRupture (SR)

Post-Gauntlet Empirical Series — August 2026

**Abstract**

This paper conducts the second empirical probe of SignalRupture (SR) as a developing cross-domain structural grammar. Electricity is selected as a hostile domain because the principal substrate is engineered rather than human: the test cannot be rescued by appealing to burnout, household distress, or psychological exhaustion. The frozen grammar is translated without changing its relational roles: effective requirement (Q) is reliability-conditioned electricity demand; durable capacity (D) is dependable deliverable resource and network capacity; sustainable flexibility (C\_S) consists of planned reserves, storage, demand response, firm imports, and other normal reliability mechanisms; extraordinary compensation (C\_E) consists only of measures outside ordinary planning assumptions; and observable function (F) is reliable delivery. The strongest available 2026 baseline is prospective rather than a completed failure sequence. NERC's 2025 Long-Term Reliability Assessment projects more than 224 GW of additional North American summer peak demand over the coming decade, a forecast 69% larger than the prior assessment's ten-year growth forecast, with data centers and other large loads driving much of the increase. This establishes a measurable moving requirement and a long-lead capacity race, but it does not establish a present continental restoration deficit or compensatory inversion. The electricity test therefore produces a scientifically important partial result: the SR grammar translates coherently into an engineered system without importing human-buffer concepts, and its Q–D timing structure is empirically measurable, but the current continental evidence is insufficient to confirm the stronger compensation-to-discontinuity sequence. The result constrains rather than inflates the metatheory.

Keywords: SignalRupture; structural grammar; electricity reliability; resource adequacy; demand growth; data centers; durable capacity; flexibility; compensatory continuity; NERC.

## 1. Why Electricity Is the Hostile Test

Healthcare can appear favorable to SR because extraordinary human labour is intuitive: workers can take extra shifts, agencies can supply temporary staff, and exhaustion is plausible. A cross-domain structural grammar cannot depend on that intuition. Electricity therefore tests whether the same relational positions survive when the system's compensatory substrate consists primarily of reserves, storage, imports, demand response, operating procedures, and emergency actions.

The test is intentionally asymmetric. SR is not permitted to relabel normal reserve engineering as hidden collapse. If the grammar requires that move, the cross-domain claim fails.

## 2. Frozen Electricity Translation

| SR role | Frozen electricity operationalization                                                                                                                                      | Classification rule                                       |
|---------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------|
| Q       | Realized/forecast peak demand plus the capacity required by the applicable reliability criterion                                                                           | Must preserve the region's technical planning standard.   |
| D       | Dependable/accredited generation and deliverable network capacity available under planning assumptions                                                                     | Nameplate capacity alone is insufficient.                 |
| C_S     | Planned reserve, normal demand response, storage, routine firm imports, operating flexibility                                                                              | Normal reliability engineering; not evidence of rupture.  |
| C_E     | Emergency imports, exceptional demand response, conservation appeals, temporary waivers, emergency reserve actions or load-shedding avoidance outside ordinary assumptions | Must be identified as extraordinary in the source regime. |

|          |                                                                                                     |                                                                                                 |
|----------|-----------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------|
| F        | Reliable electricity delivery                                                                       | Adequacy, scarcity, emergency events, involuntary curtailment, unserved energy where available. |
| $\tau_R$ | Lead time for durable generation, transmission, storage, interconnection or demand-side restoration | Domain-specific and project-specific.                                                           |

### 3. Frozen Hypotheses

| ID | Hypothesis                                                                                                                                           |
|----|------------------------------------------------------------------------------------------------------------------------------------------------------|
| E1 | As the reliability-conditioned Q–D margin tightens, dependence on compensatory mechanisms should rise under comparable stress.                       |
| E2 | Normal C_S can preserve F without constituting structural failure.                                                                                   |
| E3 | Extraordinary C_E can temporarily preserve F when ordinary capacity/flexibility is insufficient.                                                     |
| E4 | Persistent requirement growth faster than durable additions should increase the frequency or severity of extraordinary intervention, all else equal. |
| E5 | Verified durable additions or durable demand reduction that restore the margin should reduce extraordinary dependence while preserving F.            |
| E6 | If C_E becomes constrained while the Q–D gap persists, functional discontinuity risk should increase.                                                |

### 4. Baseline Evidence: A Rapidly Moving Requirement

The central empirical fact entering the frozen test is the acceleration of forecast electricity requirement. NERC's 2025 Long-Term Reliability Assessment forecasts more than 224 GW of additional North American summer peak demand over the next decade. NERC reports that this ten-year growth forecast is 69% higher than in the preceding assessment and identifies new data centers supporting AI and the digital economy as accounting for most of the projected increase, alongside industrial and electrification loads.

$\Delta Q_{\text{forecast}} > 0$  and the forecast slope has been revised upward

This is not evidence of impending grid collapse. It is evidence that Q is moving rapidly enough that a static capacity denominator is scientifically inappropriate. The relevant structural object is the trajectory of requirement relative to deliverable durable additions.

## **5. The Requirement–Capacity Race**

$$G_{C,t} = Q_t - D_t$$

$$dG_{C}/dt = dQ/dt - dD/dt$$

For electricity, this derivative is more informative than a rhetorical claim about 'stress.' If dependable capacity grows at least as quickly as reliability-conditioned requirement, the structural gap need not widen even under unprecedented load growth. If capacity additions lag realized requirement, the margin tightens. SR therefore makes a symmetric prediction: rapid demand growth can either be absorbed through successful restoration or expose temporal insufficiency.

## **6. Why Nameplate Capacity Is Rejected**

The frozen protocol defines D as dependable and deliverable capacity, not gross installed megawatts. Variable resources, transmission constraints, fuel limitations, forced-outage risk, interconnection limits, storage duration, and seasonal accreditation can make nameplate capacity a misleading denominator. This is not an SR discovery; it is standard reliability practice. SR adopts the domain's mature measurement rather than replacing it.

This is an important hostile-test result. The structural grammar survives only by respecting domain-specific semantics: the relational role is invariant, while the physical measurement is not.

## **7. Sustainable Flexibility Is Not Extraordinary Compensation**

Electric systems are intentionally designed with reserve and flexibility. Planning reserves, ordinary operating reserves, scheduled demand response, storage dispatch, and firm transfers are not evidence that durable capacity has failed. They belong to C<sub>S</sub> when they are renewable and embedded in normal reliability assumptions.

$$D + C\_S \geq Q \rightarrow \text{ordinary resilient operation}$$

This classification prevents SR from turning normal system engineering into a collapse signal. It also creates a meaningful empirical threshold: C\_E begins only when the system must move outside the normal planning envelope.

## 8. Extraordinary Compensation in an Engineered System

The electricity analogue of extraordinary human compensation is not fatigue. It is exceptional operational substitution: emergency imports, extraordinary demand reduction, conservation appeals, temporary rule changes, emergency reserve deployment, or other measures used because ordinary planned resources are insufficient under realized conditions. Such actions can be highly effective. Their presence is therefore not failure.

$$Q > D + C\_S \text{ and } C\_E > 0 \text{ while } F \text{ is maintained} \rightarrow \text{candidate Compensated Continuity}$$

The word candidate is essential. A short-lived emergency caused by an extreme event can be normal resilience. Repeated or structurally necessary C\_E under an unresolved margin is the stronger SR configuration.

## 9. Temporal Sufficiency

Electricity gives SR a cleaner version of the restoration-time problem than healthcare. Large generation and transmission projects can require years from planning through permitting, interconnection, construction, and commissioning, while load can materialize faster. The frozen timing variable is:

$$CW\_R = \tau\_buffer - \tau\_R$$

Here  $\tau\_buffer$  is not a human endurance estimate. It is the period over which existing reserve, flexibility, temporary substitution, and manageable demand can bridge the requirement before reliability criteria are breached. A negative modeled value is a planning warning, not a countdown to blackout.

## 10. AI as a Symmetric Structural Variable

The electricity test also constrains SR's treatment of AI. Data-center growth can increase Q, but AI and digital technologies can also improve forecasting, demand management, maintenance, scheduling, and operational coordination. The grammar therefore freezes a symmetric treatment:

$$\text{Net Restoration Contribution}_{\text{AI}} = \Delta D_{\text{eff, AI}} + \Delta C_{\text{S, AI}} - \Delta Q_{\text{AI}}$$

If AI-enabled capacity and flexibility outpace AI-related requirement, AI weakens the warning. If realized requirement grows faster, it strengthens the requirement–capacity race. The framework cannot classify AI as destabilizing by definition.

## 11. What the Current Evidence Actually Establishes

| Test element                                                | Current status                      | Interpretation                                                     |
|-------------------------------------------------------------|-------------------------------------|--------------------------------------------------------------------|
| Moving Q                                                    | Observed strongly                   | NERC forecast requirement has accelerated materially.              |
| Long-lead D                                                 | Established domain characteristic   | Durable additions have non-trivial development/commissioning time. |
| Q–D race measurable                                         | Yes                                 | Regional adequacy frameworks already measure it.                   |
| C <sub>S</sub> measurable                                   | Yes                                 | Reserve/flexibility constructs are mature.                         |
| C <sub>E</sub> universally harmonized                       | No                                  | Emergency classifications differ across regions.                   |
| Persistent continental $G_C > 0$                            | Not established                     | Cannot infer from demand forecast alone.                           |
| Compensated Continuity sequence                             | Not yet demonstrated continent-wide | Requires matched regional event data.                              |
| Compensatory Inversion                                      | Not demonstrated                    | Prospective.                                                       |
| Functional discontinuity caused by constrained compensation | Not demonstrated                    | Prospective/region-specific.                                       |

## 12. The First Important Negative Result

The hostile test rejects a simplistic translation of the healthcare finding. High reserve use, storage dispatch, imports, or demand response cannot automatically be interpreted as

compensation depletion. Much of this activity is the intended architecture of a reliable grid. Consequently, the human-buffer version of SR does not generalize literally to electricity.

What survives is the deeper grammar:

Requirement ↔ Durable Capacity ↔ Renewable Flexibility ↔ Exceptional  
Substitution ↔ Function

This is a useful constraint on the metatheory. Cross-domain invariance resides in relational positions, not in identical mechanisms.

### 13. Comparison With Resource-Adequacy Science

NERC, regional reliability entities, system operators, utilities, regulators, and power-system researchers already analyze adequacy, reserve margins, deliverability, forecast error, extreme events, and resource lead times. SR does not supersede these methods. The empirical burden is therefore incremental: does the frozen distinction between sustainable flexibility and exceptional compensation, combined with the timing of durable restoration, improve prospective classification or communication beyond established adequacy models?

If not, the electricity application should be treated as a translation of established reliability concepts rather than evidence of independent SR value.

### 14. Frozen Regional Panel Design

The decisive electricity study requires a region-period panel rather than a continental narrative. For each balancing authority, ISO/RTO, or reliability region where harmonization is defensible, the following variables should be assembled before outcome scoring:

| Variable | Required measure                                                             |
|----------|------------------------------------------------------------------------------|
| Q        | Realized peak; weather-normalized peak; forecast peak; planning requirement. |
| D        | Accredited dependable capacity and deliverable firm imports.                 |

|          |                                                                                                                 |
|----------|-----------------------------------------------------------------------------------------------------------------|
| C_S      | Planning reserve/flexibility already counted under ordinary criteria.                                           |
| C_E      | Predefined emergency actions outside normal assumptions.                                                        |
| F        | Emergency level, scarcity, involuntary curtailment/unserved energy, reliability-standard outcome.               |
| $\tau_R$ | Expected and realized lead time for major durable additions.                                                    |
| Z        | Weather, forced outages, fuel constraints, transmission constraints, forecast error, market/regulatory changes. |

## 15. Statistical Specification

$$M0: F_{(i,t+h)} = f(F_{(i,t)}, \text{ReserveMargin}_{(i,t)}, \text{LoadForecastError}_{(i,t)}, \text{Weather}_{(i,t)}, \text{Outages}_{(i,t)}, Z_{(i,t)})$$

$$M1: M0 + G_C(i,t) + C_E(i,t) + \Delta C_E(i,t) + \tau_R(i,t) + [G_C \times C_E]_{(i,t)}$$

M0 represents a credible electricity-sector baseline. M1 is the SR augmentation. The first criterion is not statistical significance alone but whether M1 improves out-of-sample calibration, discrimination, or forecast error. A strong result requires improvement after the best available sector variables are included.

## 16. Restoration Branch

The electricity domain makes the positive branch of SR especially important. Suppose a region faces rapid load growth, commissions sufficient dependable resources and transmission, retains adequate reserve, and experiences no increasing reliance on emergency actions. SR must classify that outcome as successful restoration.

$$dD/dt \geq dQ/dt + \text{adequate } C_S \rightarrow G_C \text{ stable/falling} \rightarrow C_E \text{ stable/falling} \rightarrow F \text{ maintained}$$

A framework that interprets successful capacity expansion as delayed collapse would fail its own frozen protocol.

## 17. Failure Branch



$dQ/dt > dD/dt \rightarrow$  margin contraction

margin contraction + recurrent C\_E  $\rightarrow$  Compensated Continuity candidate

persistent gap + constrained C\_E + F $\downarrow$   $\rightarrow$  functional discontinuity candidate

Each arrow is empirically contingent. The sequence cannot be inferred from the first line alone.

## 18. Negative Controls

- High demand growth with faster dependable capacity growth and stable reliability.
- Frequent storage/demand-response dispatch that is normal, renewable, and already included in adequacy planning.
- Emergency event caused by an isolated extreme shock followed by rapid full recovery and no residual capacity gap.
- Low reserve margin without extraordinary intervention or functional deterioration.
- High gross installed capacity but inadequate deliverability, demonstrating why nameplate capacity is not D.

## 19. Falsification Rules

- If the SR variables add no predictive or decision value beyond mature resource-adequacy models, incremental electricity-domain support fails.
- If C\_E classifications cannot be harmonized without post-hoc judgment, the compensation component is not validated in this domain.
- If regions with comparable margins show no systematic relationship between exceptional substitution and later reliability outcomes, E3–E6 weaken.
- If durable restoration closes the measured gap but SR continues to classify the region as structurally deficient, the measurement architecture is falsified.
- If the grammar requires redefining normal reserve as extraordinary compensation to obtain confirmation, the cross-domain translation fails.

## 20. Empirical Result

### Frozen proposition

Q can be defined as a moving requirement

D must be separated from gross/nameplate capacity

C\_S must be distinguished from C\_E

Compensated Continuity exists as a testable engineered configuration

Compensatory Inversion predicts discontinuity

SR adds predictive value beyond adequacy models

Grammar translates without human exhaustion

### Electricity result

Supported.

Supported, but inherited from established reliability science.

Supported conceptually and operationally; harmonization still required.

Supported as an operational possibility; not yet confirmed continent-wide.

Indeterminate.

Indeterminate; decisive statistical test not yet run.

Supported.

## 21. Cross-Domain Comparison With Healthcare

| Structural role     | Healthcare                                | Electricity                                    | Invariant?                       |
|---------------------|-------------------------------------------|------------------------------------------------|----------------------------------|
| Q                   | Service/workforce requirement             | Reliability-conditioned demand                 | Yes: effective requirement       |
| D                   | Regular sustainable staffing              | Dependable deliverable capacity                | Yes: durable capacity            |
| C_S                 | Renewable staffing flexibility            | Normal reserves/flexibility                    | Yes                              |
| C_E                 | Extraordinary overtime/purchased staffing | Emergency substitution outside normal planning | Yes in role; different mechanism |
| F                   | Volume/access/continuity                  | Reliable delivered electricity                 | Yes                              |
| Depletion mechanism | Human reserve/replacement constraint      | Resource/flexibility/temporal constraint       | No literal identity              |
| Restoration         | Durable staffing/access recovery          | Commissioned dependable capacity/flexibility   | Yes in role                      |

This is the main result of Paper II. The same relational positions can be specified in a human-intensive service and an engineered infrastructure without claiming that their mechanisms are physically identical. That is the form of invariance expected from a structural grammar.

## **22. Does Paper II Confirm SR as Structural Grammar?**

Not by itself. Paper II provides a stronger result than mere verbal analogy but a weaker result than predictive validation. The frozen roles survive translation into a hostile domain, and the requirement–capacity race is directly compatible with mature reliability measurement. However, because the current test has not yet demonstrated incremental predictive performance or a completed C\_E-to-discontinuity sequence, the correct classification is cross-domain structural compatibility, not metatheoretical confirmation.

Healthcare empirical foothold + Electricity hostile-domain compatibility ≠ final  
validation

The importance of the result is that electricity did not force SR to abandon its core relational grammar. It did force SR to narrow one interpretation: compensation is not synonymous with human depletion, and ordinary engineered flexibility must not be pathologized.

## **23. What We Have After Two Papers**

The first two empirical papers now identify a candidate invariant:

Changing Requirement → Durable Capacity Relation → Buffer/Flexibility Use →  
Observable Function

Healthcare supplies observed compensation expansion alongside continuing function. Electricity demonstrates that the same positions can be operationalized without human-buffer assumptions and exposes a prospective requirement–capacity race under rapidly increasing forecast demand. The evidence is therefore accumulating at the level appropriate to a structural grammar: relational recurrence with domain-specific mechanisms.

What remains missing is the decisive criterion: replicated incremental predictive value.

## **24. Next Cross-Domain Requirement**

The third test must be harder in a different way. Households provide heterogeneous actors, endogenous adaptation, debt and savings, varying family structure, and contested definitions of minimum requirement. If the grammar survives that domain without treating all borrowing or hardship as compensation, the case for cross-domain structural invariance becomes stronger. If it cannot, SR's scope should narrow toward organizational and infrastructural systems.

## **25. Conclusion**

The electricity hostile-domain test does not confirm a coming North American grid collapse. It confirms something narrower and methodologically more useful: the SR structural positions can be translated into an engineered reliability system without importing the human mechanism that made the healthcare case intuitively favorable. Requirement, durable capacity, sustainable flexibility, exceptional substitution, restoration lead time, and observable function are separable empirical objects.

NERC's sharply increased ten-year demand forecast makes the requirement–capacity race prospectively important, but the outcome is open. Rapid durable capacity creation, transmission expansion, storage, demand response, efficiency, load flexibility, or delayed/cancelled large loads can close the margin. Conversely, realized demand that outruns deliverable additions can increase reliance on extraordinary measures. SR must score either branch honestly.

After two domain tests, the Structural Grammar has survived translation. It has not yet earned predictive metatheoretical validation. The next scientific step is not to declare victory; it is to assemble the regional panel that can determine whether the SR augmentation predicts reliability outcomes better than established adequacy variables, while proceeding to the household test as the third cross-domain challenge.

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## Appendix A. Frozen Electricity Registry

| ID | Window    | Prediction                                                                     | Score at August 2026             |
|----|-----------|--------------------------------------------------------------------------------|----------------------------------|
| E1 | 2026–2032 | Tighter Q–D margins increase exceptional compensation under comparable stress. | Indeterminate                    |
| E2 | 2026–2032 | Normal sustainable flexibility preserves function without implying failure.    | Supported as domain architecture |
| E3 | 2026–2032 | C_E temporarily preserves F during insufficiency episodes.                     | To be tested regionally          |
| E4 | 2026–2032 | Persistent Q growth faster than D increases extraordinary intervention.        | Prospective                      |
| E5 | 2026–2032 | Durable restoration reduces extraordinary dependence.                          | Prospective                      |
| E6 | 2029–2034 | Persistent gap plus constrained C_E increases discontinuity risk.              | Prospective                      |

## Appendix B. Cross-Domain Score After Paper II

| Criterion | Status |
|-----------|--------|
|-----------|--------|

|                                               |                                |
|-----------------------------------------------|--------------------------------|
| Healthcare operationalization                 | Passed                         |
| Healthcare descriptive structural relation    | Passed / partial causal status |
| Electricity hostile-domain operationalization | Passed                         |
| Literal human-buffer universality             | Rejected — correctly narrowed  |
| Relational grammar across two domains         | Supported provisionally        |
| Incremental predictive value                  | Not yet established            |
| Three-domain minimum                          | Not yet met                    |
| Metatheory empirical confirmation             | Not yet earned                 |

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